Project Title

Stock Price Prediction with Sentiment Analysis

Submitted by

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1. Abstract

The stock market is influenced by various factors, including public sentiment expressed through social media platforms. With the growing impact of retail investors and real-time information sharing, Twitter has emerged as a significant source of market sentiment. This project explores the integration of Twitter-based sentiment analysis with time series forecasting to predict the stock price of NVIDIA Corporation (NVDA).

To analyze sentiment, the RoBERTa transformer model was utilized due to its strong capabilities in understanding contextual language. Tweets related to NVIDIA were collected, preprocessed, and passed through the RoBERTa model to extract sentiment scores—categorized as positive, negative, or neutral. These sentiment signals were then aligned with daily historical stock price data for NVIDIA.

For the forecasting task, a Long Short-Term Memory (LSTM) neural network was implemented, leveraging its ability to capture temporal dependencies in sequential financial data. Two models were trained and evaluated: one based solely on NVIDIA's historical stock prices, and another combining price data with the sentiment scores derived from Twitter.

Performance was measured using metrics such as Mean Squared Error (MSE) and directional accuracy. The results demonstrated that the inclusion of sentiment data provided a modest improvement in predictive accuracy, especially during periods of heightened social media activity or market volatility.

This project illustrates the potential of combining deep learning with social media sentiment analysis for stock prediction and highlights how platforms like Twitter can offer valuable real-time signals in financial forecasting models.

2. Introduction

The stock market is a dynamic and complex system influenced by a multitude of factors ranging from macroeconomic indicators and company-specific news to investor behavior and public sentiment. In recent years, the role of sentiment—especially that expressed on social media platforms—has gained significant attention in financial forecasting and decision-making. Among these platforms, Twitter (now "X") stands out as a real-time, high-volume source of public opinion, offering insights into the collective mood and expectations of retail and institutional investors alike.

With the democratization of financial markets and the rise of retail investors, platforms like Twitter (now "X") have become instrumental in shaping market trends. Tweets reflecting optimism, fear, or skepticism can rapidly influence investor behavior and, in turn, impact stock prices.

This project aims to explore the integration of Twitter (now "X")-based sentiment analysis with time series forecasting techniques to predict the stock price movement of NVIDIA Corporation (ticker: NVDA). NVIDIA, a global leader in graphics processing units (GPUs) and AI computing, has seen increased attention from both investors and the public due to its pivotal role in the rapidly growing artificial intelligence and semiconductor sectors. Given the company's prominence and volatility, it serves as an ideal case study for examining the predictive power of sentiment-driven models.

The objective of this project is twofold: first, to extract and quantify sentiment from tweets related to NVIDIA using state-of-the-art natural language processing (NLP) techniques; and second, to integrate this sentiment data with historical stock price data in a time series forecasting model. By combining textual sentiment signals with numerical financial data, this study seeks to improve the accuracy and responsiveness of stock price predictions, ultimately contributing to more informed and timely investment strategies.

3. Related Work

- Ray et al. (2021) introduced a hybrid BST-LSTM model that incorporates news sentiment into stock market forecasting. By combining BST's probabilistic modeling with LSTM's nonlinear learning capabilities, the model effectively identifies anomalies and captures market trends. It demonstrated superior short-term prediction accuracy with lower MSE and MAPE than traditional models [1].

- Alam et al. (2024) introduced a hybrid LSTM-DNN model that enhances stock market prediction by combining the strengths of both architectures. The model demonstrates strong adaptability, accuracy, and resilience across diverse datasets, showing significant improvements in predictive performance compared to existing methods. Its ability to capture longer-term trends and outperform traditional approaches highlights its potential for guiding profitable investment decisions [2].
- Ren et al. (2018) investigated the role of investor sentiment in forecasting stock market direction by extracting sentiment indexes from financial review texts on platforms like Sina Finance and Eastmoney. They adjusted these indexes for anomalies such as the day-of-week effect and employed an SVM model to predict the SSE 50 Index. Their approach achieved a high prediction accuracy of 89.93%, showing that integrating sentiment data with market data significantly enhances forecasting performance and supports better investment decisions [3].
- Minh et al. (2018) proposed a Two-stream Gated Recurrent Unit (TGRU) framework for stock trend prediction, combining sentiment-aware Stock2Vec embeddings with key financial indicators. The model, which learns in both forward and backward directions, outperformed traditional GRU and LSTM architectures with a 66.32% accuracy. It proved effective on both the S&P 500 and individual stocks, and simulations showed its robustness against market volatility and potential for integration into automated trading systems [4].
- Bacco et al. (2024) analyzed predictive modeling for bank stock prices in the US and Europe, revealing that regional economic dynamics strongly influence model effectiveness. Their findings showed that individual bank-based models work better in the US due to higher inter-bank correlations, while aggregated data models perform better in Europe due to diverse national policies. Incorporating Twitter sentiment improved forecasts, particularly for North American banks, and recent data yielded more accurate predictions. The study underscores the value

of region-specific model tuning and multi-source data integration in financial forecasting [5].

- Zhang et al. (2024) proposed the TK-GAN model for stock price prediction, which integrates sentiment factors, a Soft Attention mechanism, BERT pre-trained models, and the AdamK optimizer. Evaluated on the Guizhou Maotai stock dataset, the model achieved lower error values compared to baseline approaches. While effective with textual data, the study suggests incorporating additional financial statement data and volatility charts in future work to enhance cross-modal learning and overall prediction accuracy [6].
- Li et al. (2018) introduced a sentiment transfer learning framework to enhance stock prediction performance for stocks with limited financial news. The approach transfers knowledge from news-rich stocks to news-poor ones using three source selection principles—price correlation, sector similarity, and validation performance—combined through a majority voting mechanism. Experimental results show that this method significantly outperforms non-transfer baselines and single-principle selection strategies, demonstrating its robustness in real-world stock prediction scenarios [7].

4. Methodology and Implementation

This project explores a data-driven pipeline for predicting NVIDIA's stock price by integrating social media sentiment with time-series forecasting. The process began with the collection of historical stock data for NVIDIA (NASDAQ: NVDA) from Yahoo Finance for the period of Q1 2024. In parallel, relevant tweets were scraped using the twscrape library, leveraging asynchronous API calls to collect high-volume data over several weeks.

The raw tweet data was cleaned through a series of preprocessing steps including lowercasing, link and emoji removal, tokenization, stopword filtering, and lemmatization. Sentiment classification was performed using the RoBERTa model, which had been pre-trained on over 124 million tweets. RoBERTa (A Robustly Optimized BERT Pretraining Approach) is a transformer-based model that modifies BERT by removing the next sentence prediction task and dynamically changing the masking pattern applied to the training data. It consists of multiple self-attention layers, and uses a bidirectional encoder to understand the full context of a word based on its surroundings.

To classify sentiment, the model uses the softmax function:

$$Softmax(z_i) = e^{z_i} / \sum_{j=1}^{n} e^{z_j}$$

This converts the model's raw output scores (logits) into a probability distribution over the sentiment classes: positive, neutral, and negative. The predicted class corresponds to the highest probability. Sentiment labels :- positive, neutral, or negative, were mapped to numeric values 1, 0, and -1, respectively, and aggregated by date.

The sentiment data was then merged with the daily stock data to create a unified dataset containing features like opening price, closing price, trading volume, average daily sentiment, and tweet volume. This dataset served as input for an LSTM (Long Short-Term Memory) neural network. LSTMs are a type of Recurrent Neural Network (RNN) capable of learning long-term dependencies through the use of gates: input, forget, and output gates. These gates control the flow of information and help maintain gradient stability over long sequences. Mathematically, the LSTM updates are:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\overline{C_t} = tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

$$C_t = f_t * C_{t-1} + i_t * C_t$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t * tanh(C_t)$$

Two separate LSTM models were developed and trained: a baseline model using only traditional stock indicators, and a second model incorporating Twitter-derived sentiment features alongside stock data.

The dataset was split into training and testing subsets with an 80:20 ratio. Both models were implemented using PyTorch and trained for 512 epochs. The input data was normalized using MinMax scaling, and five-day time windows were used to predict the stock movement for the next day. The performance was evaluated using Mean Squared Error (MSE) as the loss function:

$$MSE = 1/n \sum_{i=1}^{n} (y_i - \widehat{y_i})^2$$

Additionally, the correlation between stock metrics and sentiment indicators was assessed using the Pearson Correlation Coefficient, given by:

$$r = [\Sigma\{(x_i - \overline{x})(y_i - \overline{y})\}] \,/\, sqrt[\Sigma(x_i - \overline{x})^2 \cdot \Sigma(y_i - \overline{y})^2]$$

Performance comparison between the two models provided insights into the role of sentiment in financial forecasting.

5. Graphs and Images

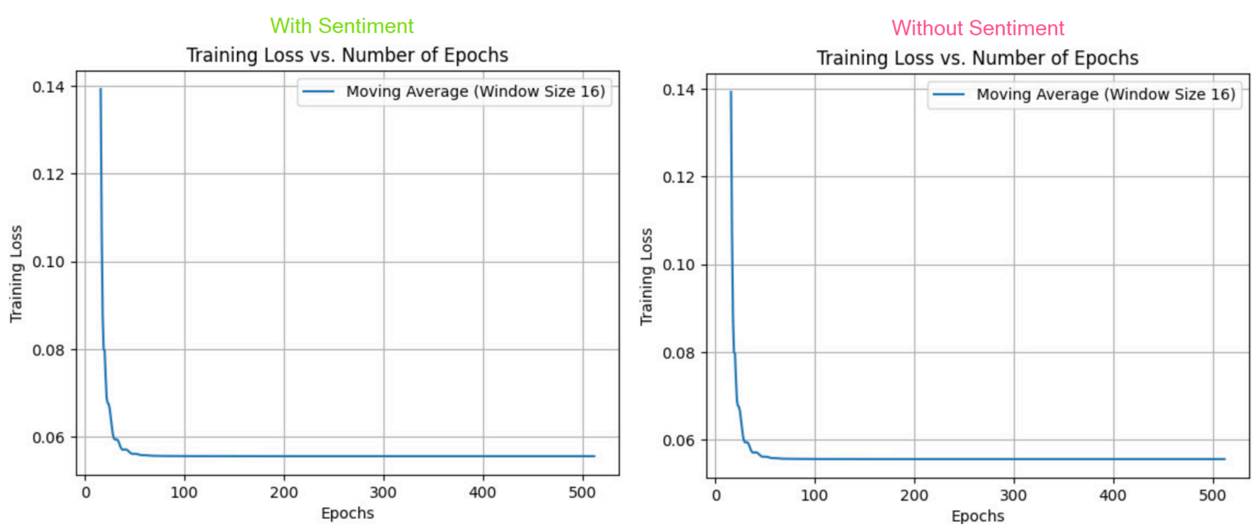

Fig 1: Training Loss vs Number of Epochs

This Fig 1 presents a side-by-side comparison of training loss over 512 epochs for two LSTM models: one trained with sentiment input (left) and the other without sentiment (right). In both plots, the training loss rapidly decreases in the initial epochs and stabilizes around 0.042, as shown by the smoothed moving average curve (window size = 16). The nearly identical shape and convergence point of both curves suggest that incorporating sentiment data did not significantly impact the model's training efficiency or learning behavior. This visual comparison supports the conclusion that, under the given dataset and timeframe, sentiment information had minimal influence on the LSTM's ability to fit the training data.

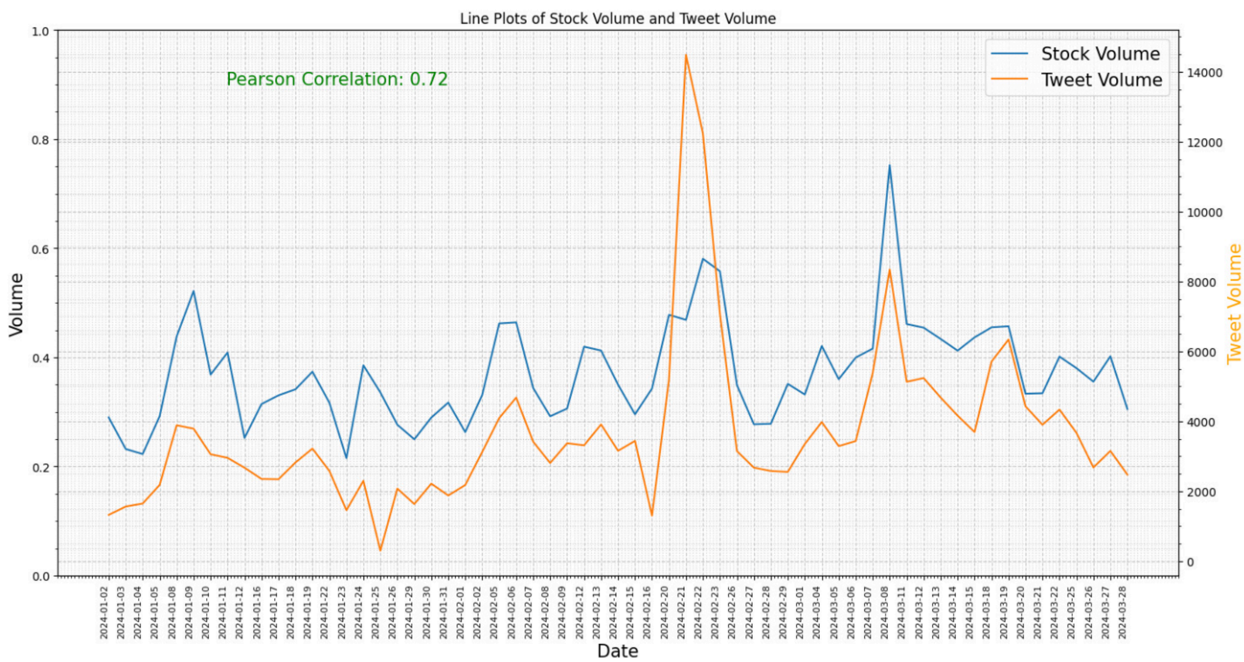

Fig 2 : Line Plot of Stock Volume of NVIDIA vs Tweet Volume

This graph in Fig 2 compares NVIDIA's daily stock trading volume (in blue) with the corresponding tweet volume (in orange) over Q1 2024. The stock volume data has been normalized for scale, while tweet volume is shown on the secondary Y-axis. Both series exhibit a strong visual alignment, especially around major spikes, suggesting periods of heightened public and market activity. The Pearson correlation coefficient of 0.72, shown in green, indicates a strong positive correlation, meaning that an increase in the number of tweets about NVIDIA generally corresponds with increased trading volume on the stock market. This supports the idea that Twitter activity may reflect real-time investor interest and could serve as a leading indicator for trading behavior.

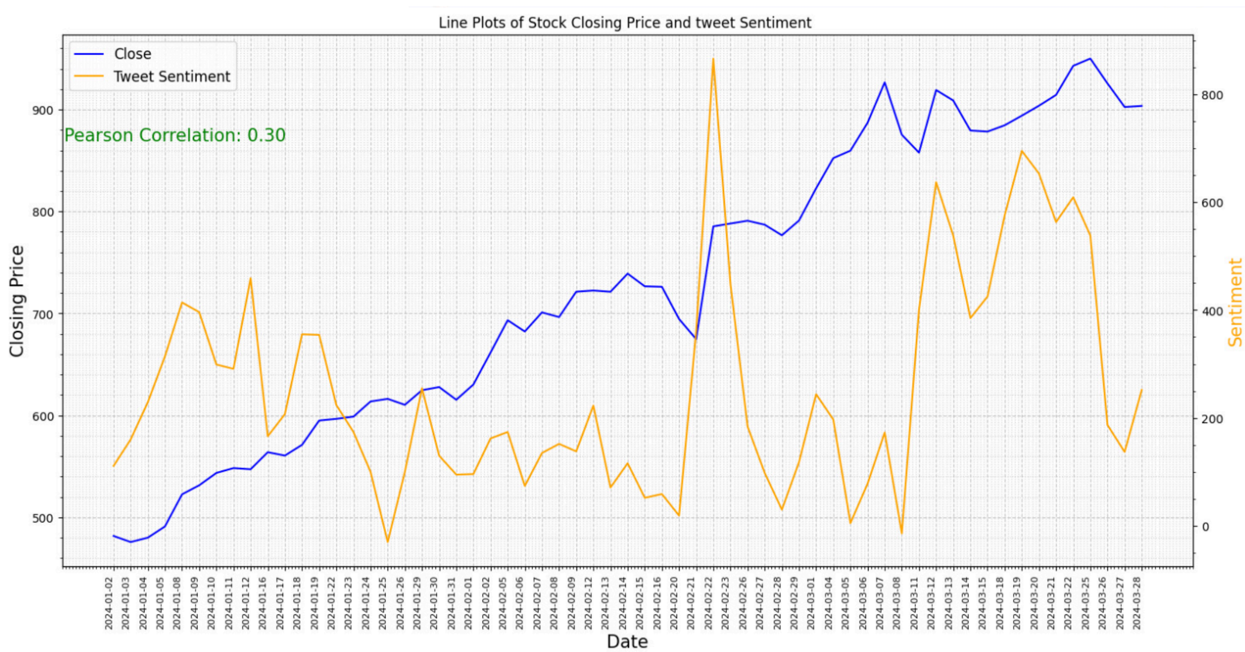

Fig 3 : Line Plot of Closing Price of NVIDIA vs Tweet Sentiments

This graph in Fig 3 illustrates the daily closing price of NVIDIA (in blue) alongside the aggregated daily tweet sentiment volume (in orange) for Q1 2024. The stock price shows a general upward trend over the period, while tweet sentiment fluctuates more sharply, indicating volatile social media activity. A Pearson correlation coefficient of 0.30 suggests a weak positive relationship between the two variables, meaning that higher tweet sentiment is mildly associated with higher closing prices, though the connection is not strong or consistent. This visual supports the idea that while social media sentiment may reflect market mood, it does not reliably predict short-term stock movements on its own.

6. Results

6.1. Model Performance

To evaluate the effectiveness of sentiment information in stock price prediction, two LSTM models were trained: one using only historical stock price data, and another incorporating sentiment scores generated by a fine-tuned RoBERTa model. Both models were evaluated using the Mean Squared Error (MSE) on the test set. Surprisingly, both models achieved the same final MSE, suggesting that while sentiment data may influence price trends in specific scenarios, it did not significantly improve the overall predictive accuracy across the dataset.
Additionally, it was observed in Fig 1 that after approximately 100 to 200 epochs, the MSE loss plateaued and showed no significant change. This indicates that the models had converged and additional training beyond this point did not yield further improvements in performance.

6.2. Correlation Analysis

To understand the relationship between public sentiment and market behavior, Pearson correlation coefficients were computed for relevant variable pairs:

- Stock Price and Net Sentiment: A weak positive correlation of 0.3 was observed as apparent in Fig 3, indicating a limited linear relationship. However, specific instances suggest sentiment may influence price movement. For example, between 20th February 2024 and 22nd February 2024, a sharp rise in net sentiment coincided with a substantial increase in stock price, hinting at a possible short-term impact of sentiment on market behavior.
- Stock Volume and Tweet Volume: A much stronger correlation of 0.72 was found, showing that trading activity and public discussion volume are closely linked. This relationship is visually confirmed in Fig 2, where stock volume and tweet volume follow similar trends, suggesting tweet frequency may be a strong indicator of trading interest or market attention.

6.3. Observations

Although the inclusion of sentiment data did not improve MSE performance quantitatively, correlation analysis and case-specific trends reveal that sentiment may still provide valuable signals in certain market conditions. Thus, sentiment analysis can complement traditional price prediction models by offering additional context, especially for sudden market movements.

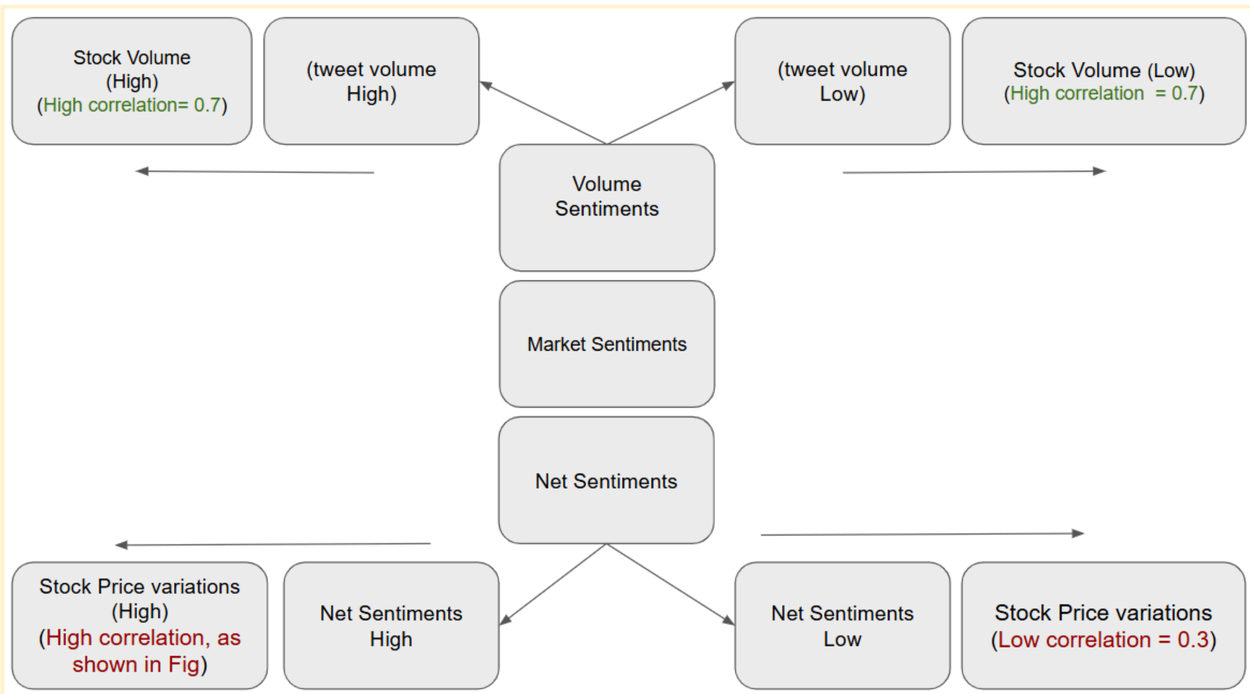

Fig 4 : block diagram of volume and price behavior with respect to tweet volume and sentiments

7. Conclusion:

The project, titled “Stock Price Prediction Using Sentiment Analysis”, explored the integration of sentiment analysis with traditional time-series forecasting for stock price prediction. A fine-tuned RoBERTa model was employed to extract sentiment scores from financial tweets, which were then used alongside historical stock data to train LSTM-based predictive models. The goal was to assess whether sentiment information could enhance prediction accuracy.

While the inclusion of sentiment data did not improve the model's quantitative performance—as both LSTM models with and without sentiment input achieved the same MSE on the test set—the qualitative analysis revealed that sentiment can influence market behavior in specific instances. Notably, a spike in net sentiment corresponded with a sharp price increase during the observed period of 20–22 February 2024 for NVIDIA (NVDA) stock. Furthermore, the strong correlation (0.72) between tweet volume and stock trading volume highlights the potential of social media activity as a signal of market interest.

Overall, the findings suggest that although sentiment data alone may not drastically improve predictive accuracy in a general sense, it holds value as a supplementary feature that can offer insight into short-term price dynamics and investor behavior. Future work can focus on combining sentiment with additional market indicators or exploring advanced architectures that better capture complex interactions between textual sentiment and market movements.

8. Future work :

future work may involve:

- Expanding the dataset to cover longer timeframes (e.g., one year or more).
- filtering out low-sentiment volumes as noise
- Testing more complex or alternative models such as GRU, vanilla RNN or new GAN based neural networks.
- Additionally, the training process can be optimized by reducing the number of epochs from 500 to 100–200, as performance gains plateau beyond this range.

9. References :

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